

Identifying Firm Characteristics for Successful Co-Creation - Literature Review

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Abstract. In the last decade, active involvement of end-users in new product development (NPD) commonly referred as co-creation has emerged as an effective strategy to overcome barriers that hinder firms from understanding customer needs. In response, co-creation has generated a great deal of interest for both practitioners and researchers in different areas of NPD. Despite co-creations growing trend, relatively little is known about this new form of product development. Hence this study provides a ready reference of recent literature on co-creation. First, this study distinguishes between the roles and responsibilities of firm and end-users at different phases of NPD during traditional and co-creation projects. Second, after an extensive literature review of marketing, innovation management and product development it identified important managerial characteristics needed by the firms to successfully implement co-creation.

Keywords. Co-creation, new product development, customer involvement, characteristics, success factors, firm characteristics

Introduction

Co-creation or active involvement of end-users during New Product Development (NPD) is not a new concept. It was first introduced in Europe in 1970s and was referred as “participatory design” [1]. Ever since, researchers and practitioners alike have tried to utilize the concept to minimize the product failure rate and have given it different names such as, co-designing [2], [3], design by customer [4], participatory design [5], [6], user generated content [7], mass collaboration [8], [9], co-production [10] customer participation [11], co-creation [9], [12]–[14] and so on. Despite the name given all these methods emphasize on the same thing that is end-users involvement in the process of NPD.

In the last decade, the concept has been generally termed as co-creation and received a large amount of attention from both practitioners and researchers to minimize the risk of misinterpretation of end-users needs during NPD. This rise in the use of the concept has caused the literature to grow and not only many folds but in many directions. This breadth of literature has created difficulties for the new researchers and practitioners beginning to work in the field. This study with the help of literature published in the fields of marketing, innovation management and product development identified important managerial characteristics needed by the firms to successfully implement co-creation.

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Along with the breadth of literature the use of term co-creation in many contexts has led to confusion, where it is argued that all NPD projects (traditional or co-creation) heavily rely on end-users input and involve them in the process. But involvement during traditional NPD is passive and end-users only provide feedback when they are asked [15], whereas emphasize during co-creation NPD is on active end-users involvement where firms assume the role of mere supporters during many phases of the project. Therefore, co-creation can be defined as an active, creative and social process which includes some special methods and strategies applied by firms to engage the end-users into the firms' initiated NPD projects [13], [16], [17]. To clear this confusion following section is dedicated to distinguish between the roles of end-users during traditional and co-creation NPD.

1. Role of end-user in traditional and co-creation NPD projects

The main idea of co-creation is to involve its end-users into the process of NPD. The end-users involvement in co-creation projects not only differs in points of interactions, but also in intensity, role and goal from these interactions when compared to traditional NPD projects. For the traditional NPD projects as shown in Figure 1, the role of the end-users is supportive. All the primary functions /decisions are made by firms, and the end-users only provide feedback when they are asked [15]. Concepts/ideas are generated by the firms and after that the end-users feedback is obtained through surveys or interviews to verify and validate the generated concept [18], [19]. Often the surveys and interviews are designed to support the concepts generated by the firms and only a few adjustments based on the feedback obtained are made and the concepts are finalized by the firms. All important activities of designing, developing and prototyping are conducted solely by the firms and are governed by price, quality and technical capabilities of the firms [19]. After the development of prototypes, the end-users are asked to provide the feedback on the developing products [20], [21]. Few changes are possible at this point as most of the product development work is already completed and it is very difficult for the firms to make any major changes in the designs due to competitive market pressure to launch the products [22],[23]. Based on the obtained feedback, the firms refine the products and prepare for full production. No further end-users involvement is considered until the products are available in the market. The end-users are then asked to provide feedback on the products offered for the concept generation of the future offerings [12].

Contradictory to the traditional NPD projects, the role of the end-users in co-creation NPD projects at many phases is also primary [1], [13], [24], [25], as shown in Figure 2. The ideas/concepts are mainly generated by the end-users while the firms only provide support/platform for the end-users to use their creativity to find solutions for the problems presented to them [1], [26], [27]. Once the ideas are generated, the firms usually refine the ideas through the evaluation, on the basis of feasibility and technical capabilities of the firms [1], [24]. After the concepts are refined, the end-users and firms again work together to finalize the concepts. All the other remaining phases of the NPD projects completed by the collaboration of the firms and end-users until the products are ready to be launched into the market [13], [28]. In the co-creation NPD projects, the end-users not only contribute to the manufacturing of the successful products but also to promote the products to the market at little or no cost for the firms [9], [29].

In short, it can be concluded that the NPD projects that only take the feedback from the end-users, provide multiple options to choose from the offered variations or even try to tap the current experiences of the users for future offerings are not the co-creation NPD projects [12]. In contrast, the projects in which the firms facilitate the involvement of the end-users into the process of NPD where end-users lead the way for the firms to act according to their wishes can be classified as the co-creation projects [1], [12], [30]. For example, a firm offering its end-users colorless cars with tools and platform to design on their own. This activity may result in a normal car with regular color paint or may result in different types of design or graffiti on the car as happened in Daimler AG 2010 "Style Your Smart" competition. This type of projects can be classified as co-creation NPD projects. On the other hand, if a firm offers cars to its end-users and provides them multiple color options to choose from for those cars, it is not co-creation.

Traditional NPD

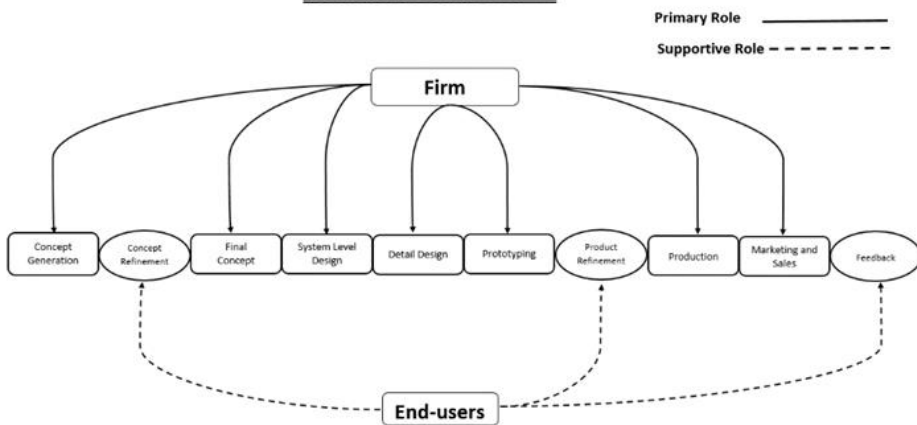


Figure 1. Role of end-users in traditional new product development process.

Co-Creation NPD

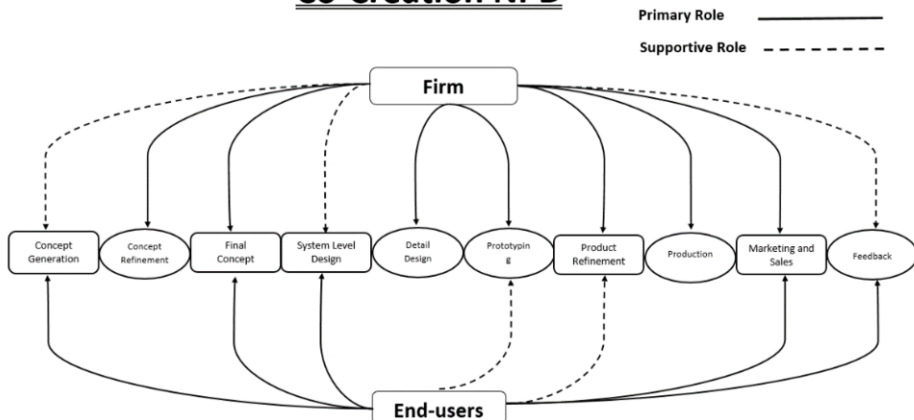


Figure 2. Role of end-users in co-creation based new product development process.

2. Firm characteristics for co-creation

In last decade co-creation has received an overwhelming response from both researchers and practitioners and literature body has grown not only many folds but in many directions. Though the work related to co-creation has been primarily published in marketing literature, yet a great number of original contributions have been published in innovation management and product development literature. This breadth of literature resulted in difficulties for new researchers and practitioners looking to employ the concept. This study attempts to identify a list of important managerial characteristics for co-creation that will serve as ready reference for new researchers and practitioners in the field.

2.1. Two-way communication channel

Co-creation is a process of engaging end-users into NPD projects and a reliable two-way communication channel is necessary for nourishing this engagement [14], [31]. Therefore, unless a communication exists between a firm and its co-creators, successful co-creation will be very difficult.

2.2. Staff training in customer relation

Irrespective of roles played, every employee who interacts with end-users is responsible for the success of co-creation [32]. According to a study conducted by Paul and Seddighi [14], on the management and operational capabilities for innovation via co-creation, the importance of staff training was highly emphasized by the managers of the best performing firms in the northeast region of England.

2.3. Manufacturing personalized items

Previous studies have confirmed that relationships between firms and end-users depend upon the types of services/products offered by the firm. High interaction personalized products (i.e. watch, smartphone, laptop) offers much more opportunities for co-creation in comparison to non-personalized items [33], [34]. That is because the end-users are more motivated to participate in such projects.

2.4. Firm willingness

Willingness is self-motivation that individual must have to initiate any activity. This is held true also for co-creation. The degree of a firm's willingness to co-create is directly proportional to the degree of co-creation the firm can achieve [34], [35]. Therefore, the willingness of the firm to co-create is a key requirement to engage in co-creation as the process needs to be initiated by the firm itself [13].

2.5. Effective information sharing

In NPD projects, appropriate/effective information sharing with end-users is very critical because this information will be available to not only end-users but competitors

also. Therefore, the extent and method for information sharing is a strategic decision [36] and should be taken carefully.

2.6. Effective management of the new content

For a firm engaged in co-creation, a large amount of information is expected to be shared from its end-users. If it is not properly managed, it will soon lead to information overload [37] and may become a challenge for the firm.

2.7. Mass customization experience

Firms already engaged in mass customization have the experience to manage end-users feedback and have already developed the concept of listening to the end-users. Thus, these firms can develop competencies required for co-creation easily compared to those with no mass customization experience [14], [34].

2.8. Communication among end-users

Communication among end-users is important [14], especially for a firm using its end-users for marketing, idea generation, and screening of ideas generated by other users. The process of using the end-users to do all these can be exploited with the help of platform provided by the firm to its co-creators. With the help of the provided platform, the co-creators not only can discuss the new ideas with each other but also can help each other in obtaining better conclusion by providing instant feedback and recommendations to the peer co-creators [38].

2.9. Exploitation

Exploitation refers to the ability of a firm to quickly convert innovative ideas into successful products for financial benefits [36]. As the ideas available on the public domains are vulnerable to be copied easily by competitors. Therefore, having ability of to quickly exploit the ideas will increase the success of co-creation projects.

2.10. Current market share

The degree of willingness of end-users to co-create determines the level of co-creation a firm can achieve [34]. It is more likely that the end-users will be motivated to co-create with the market leader or well-known brands in comparison to those with little market acceptance.

2.11. Satisfaction level

According to a study by Füller [39], one key reason for end-users to engage in co-creation was their current satisfaction level with firms' offerings. The end-users participated in the firms initiated co-creation projects to improve their experience for future offerings.

2.12. Multiple communication channels

As previously mentioned, communication between end-users and a firm is the core of co-creation, and high number of interactions will have positive impact on co-creation results [31], [38]. Therefore, it will be advantageous for the firm to have multiple communication channels to interact with the end-users during co-creation.

2.13. Screening system

Co-creation activities yield a large number of ideas that will soon lead to information overload if they are not properly managed [37] and may result in unnecessary complexity during the process. Therefore, a firm is required to have a screening system for filtering ideas given by its co-creators to save time and efforts.

2.14. Effective Information Management System (IMS)

Information Management System (IMS) is an information storing, searching and retrieval system that presents right information to right users according to their requirements and responsibilities. Information sharing within a firm is very important during co-creation projects [14] and with the help of effective IMS, the firm can distribute important information timely and efficiently.

2.15. Openness to ideas

Willingness of a firm plays a vital role in the level of co-creation it can achieve [34]. That includes not only top management but also managers and designers working on a project. Many firms face “Not Invented Here” (NIH) syndrome [39] where design teams do not accept ideas generated by anyone else. This attitude affects directly the level of willingness of the firm to co-create. Therefore, it is very important for the firm to develop a mindset to be open to ideas from everyone [35].

2.16. R&D Activities

A firm that does not engage in R&D activities cannot act according to the voice of end-users as it only produces what has already existed and well established in a market. The firm does not look into ways to fulfill latent needs of the end-users. That is why R&D activities were marked necessary for co-creation by many managers of high performance firms during investigation of firm capabilities for co-creation [14].

2.17. Flexibility

During the relevant literature review it was observed that the firm engaged in co-creation may receive novel/creative ideas from end-users at any phase of NPD and may have to incorporate them quickly into ongoing projects to be the first to offer the solution to the market and gain competitive advantage. Therefore, the firm willing to engage in co-creation should have flexibility incorporated into the existing system. But this important characteristic of the firms was missing from the literature and hence was added into the list to make it more robust and complete. Where, flexibility refers to a

firm's abilities to incorporate ideas/suggestions created by co-creators rapidly into its existing system.

These identified attributes not only reflect on the managerial capabilities of the firms, but also the mindsets of the firms and their readiness and intentions towards co-creation. It was also noted that the importance level of these characteristics of co-creation is not same and different characteristics may have different levels of impact on co-creation success. Further studies can confirm these importance levels. It is also foreseen that these identified characteristics can be used in future studies for developing methods/frameworks for successful co-creation.

3. Conclusion

Besides highlighting the shift in roles and responsibilities of both firm and end-users during NPD, seventeen important characteristics needed by firms for co-creation are identified from recent literature. During the identification of these characteristics the focus was intentionally kept on managerial approaches and mindsets of the firms. The characteristics related to technical and operational capabilities of firms were ignored as their need may vary depending upon the nature of industry and size of the firm. Therefore, it is argued that these identified characteristics are equally applicable to all industry sectors and product categories. This list of characteristics will serve as a ready reference for not only practitioners but also new researchers beginning to work in the field of co-creation by providing them highlights of what qualities/characteristics are needed from firms for successful co-creation.

References

- [1] E. B.-N. Sanders and P. J. Stappers, Co-creation and the new landscapes of design, *CoDesign*, Vol. 4, 2008, No. 1, pp. 5–18.
- [2] P. Bate and G. Robert, Experience-based design: from redesigning the system around the patient to co-designing services with the patient., *Qual. Saf. Health Care*, Vol. 15, 2006, No. 5, pp. 307–310.
- [3] E. B. Sanders, Generative Tools for Co-Designing, in *Collaborative Design*, 2000, pp. 3–12.
- [4] Risdiyono and P. Koomsap, Design by customer: Concept and applications, *J. Intell. Manuf.*, Vol. 24, 2013, No. 2, pp. 295–311.
- [5] E. Ruhl, C. Richter, J. Lembke and H. Allert, Beyond methods : Co-creation from a practice-oriented perspective, *Proc. DRS 2014*, 2014, pp. 967–979.
- [6] C. R. Wilkinson and A. De Angeli, Applying user centred and participatory design approaches to commercial product development, *Des. Stud.*, Vol. 35, 2014, No. 6, pp. 614–631.
- [7] A. Karahasanović, P. B. Brandtæg, J. Heim, M. Lüders, L. Vermeir, J. Pierson, B. Lievens, J. Vanattenhoven, and G. Jans, Co-creation and user-generated content-elderly people's user requirements, *Comput. Human Behav.*, Vol. 25, 2009, No. 3, pp. 655–678.
- [8] F. Piller, P. Schubert, M. Koch and K. M. Möslin, Overcoming Mass Confusion: Collaborative Customer Co-Design in Online Communities, *J. Comput. Commun.*, Vol. 10, 2005, No. 4, article 8.
- [9] V. Zwass, Co-Creation: Toward a Taxonomy and an Integrated Research Perspective., *Int. J. Electron. Commer.*, Vol. 15, 2010, No. 1, pp. 11–48.
- [10] M. Etgar, A descriptive model of the consumer co-production process, *J. Acad. Mark. Sci.*, Vol. 36, 2008, No. 1, pp. 97–108.
- [11] S. Nambisan and R.A. Baron, Virtual customer environments: Testing a model of voluntary participation in value co-creation activities, *J. Prod. Innov. Manag.*, Vol. 26, 2009, No. 4, pp. 388–406.
- [12] C.K. Prahalad and V. Ramaswamy, Co-Creating Unique Value With Customers, *Strateg. Leadersh.*, Vol. 32, 2004, No. 3, pp. 32–33.
- [13] C.K. Prahalad and V. Ramaswamy, Co-creation experiences: The next practice in value creation, *Journal of Interactive Marketing*, Vol. 18, 2004, No. 3, pp. 5–14.

- [14] P. Reay and H. R. Seddighi, An empirical evaluation of management and operational capabilities for innovation via co - creation, *Eur. J. Innov. Manag.*, Vol. 15, 2012, No. 2, pp. 259-275.
- [15] E. Von Hippel, A customer-active paradigm for industrial product idea generation, *Res. Policy*, Vol. 7, 1978, No. 3, pp. 240-266.
- [16] E. von Hippel, Democratizing innovation: The evolving phenomenon of user innovation, *J. für Betriebswirtschaft*, Vol. 55, 2005, No. 1, pp. 63-78.
- [17] F. T. Pillier, C. Ihl, and A. Vossen, A Typology of Customer Co-Creation in the Innovation Process, *SSRN Electron. J.* , 2010, pp. 31-61.
- [18] K. Matzler and H. H. Hinterhuber, How to make product development projects more successful by integrating Kano's model of customer satisfaction into quality function deployment, *Technovation*, Vol. 18, 1998, No. 1, pp. 25-38.
- [19] K.T. Ulrich and S.D. Eppinger, *Product Design and Development*, 4th. ed, McGraw-Hill, Boston, 2008.
- [20] N.F.M. Roozenburg and J. Eekels, *Product Design: Fundamentals and Methods*, John Wiley & Sons, Hoboken, 1995.
- [21] E.B. Sanders, Information, Inspiration and Co-creation, *6th Int. Conf. Eur. Acad. Des.*, 2005, pp. 29-31.
- [22] D. Chang and C-H. Chen, Understanding the influence of customers on product innovation, *Int. J. of Agile Systems and Management*, Vol. 7, 2014, Nos. 3/4, pp. 348-364.
- [23] S. Alguezaui and R. Filieri, A knowledge-based view of the extending enterprise for enhancing a collaborative innovation advantage, *Int. J. Agile Systems and Manag.*, Vol. 7, 2014, No. 2, pp. 116-131.
- [22] B. Jaworski and A. Kohli, Co-creating the voice of the customer, in *The service-dominant logic of marketing: dialogue, debate, and directions*, 2006, pp. 109-117.
- [23] C.K. Prahalad and V. Ramaswamy, The Co-Creation Connection, *Strateg. Bus.*, Vol. 27, 2002, No. 27, pp. 50-61.
- [24] F.T. Pillier and D. Walcher, Toolkits for idea competitions: A novel method to integrate users in new product development, *R D Manag.*, Vol. 36, 2006, No. 3, pp. 307-318.
- [25] J. Matthing, B. Sandén and B. Edvardsson, New service development: learning from and with customers, *Int. J. Serv. Ind. Manag.*, Vol. 15, 2004, No. 5, pp. 479-498.
- [26] J. Füller and K. Matzler, Virtual product experience and customer participation-A chance for customer-centred, really new products, *Technovation*, Vol. 27, 2007, No. 6-7, pp. 378-387.
- [27] M.J. Culnan, P.J. McHugh and J.I. Zubillaga, How Large U.S. Companies Can Use Twitter and Other Social Media to Gain Business Value, *MIS Q. Exec.*, Vol. 9, 2010, No. 4, pp. 243-259.
- [28] C. Lawer, On customer knowledge co-creation and dynamic capabilities, *Comput. Inf. Sci.* , 2005, No. Miscellaneous Papers, pp. 1-19.
- [29] A. Gustafsson, P. Kristensson and L. Witell, Customer co - creation in service innovation: a matter of communication?, *J. Serv. Manag.*, Vol. 23, 2012, No. 3, pp. 311-327.
- [30] V. Ramaswamy, Leading the transformation to co - creation of value, *Strateg. Leadersh.*, Vol. 37, 2009, No. 2, pp. 32-37.
- [31] H. H. Bauer, N. E. Sauer, and C. Becker, Investigating the relationship between product involvement and consumer decision-making styles, *J. Consum. Behav.*, Vol. 5, 2006, pp. 342-354.
- [32] D. Lazarus, A. Krishna and S. Dhaka, Co-creation Willingness Matrix and Capability Continuum for Classification and Scaling of Services., *J. Glob. Mark.*, Vol. 27, 2014, No. 4, pp. 213-225.
- [33] W.H. Voorberg, V.J.J. M. Bekkers and L. G. Tummers, A Systematic Review of Co-Creation and Co-Production: Embarking on the social innovation journey, *Public Manag. Rev.*, July 2014, pp. 1-25.
- [34] F. Pillier and C. Ihl, *Open Innovation with Customers*, RWTH Aachen, 2009.
- [35] W. D. Hoyer, R. Chandy, M. Dorotic, M. Krafft and S. S. Singh, Consumer Cocreation in New Product Development, *J. Serv. Res.*, Vol. 13, 2010, No. 3, pp. 283-296.
- [36] K. Verleye, The co-creation experience from the customer perspective: its measurement and determinants, *J. Serv. Manag.*, Vol. 26, 2015, No. 2, pp. 321-342.
- [37] J. Füller, Why consumers engage in virtual new product developments initiated by producers, in *Advances in Consumer Research*, 2006, vol. 33, pp. 639-646.
- [38] T. Roser, Aligning the co-creation project portfolio with company strategy, *Strateg. Leadersh.*, Vol. 42, 2014, No. 1, pp. 30-36.
- [39] R. Katz and T. J. Allen, Investigating the Not Invented Here (NIH) Syndrome, *R D Manag.*, Vol. 12, 1982, No. 1, pp. 7-19.